

2026 EDITION • ZERO DUPLICATES • UPDATED TOPICS

100 GenAI Interview Questions

From Transformers to production system design, in the order you should study them.

Transformers

Prompting

Embeddings

Vector DBs

RAG

Fine-Tuning

AI Agents

Evaluation

System Design

TOP = asked most often in senior interviews

Swipe to start →

01 · LLM & Transformer Fundamentals

Q 01-20

- 01 What is a Large Language Model (LLM)?
- 02 How does the Transformer architecture work?
- 03 Why did Transformers replace RNNs and LSTMs?
- 04 What is self-attention?
- 05 How does multi-head attention work?
- 06 What are Query, Key, and Value (Q, K, V)?
- 07 What is positional encoding (and RoPE), and why is it needed?
- 08 Encoder-only vs decoder-only vs encoder-decoder models?
- 09 Why are most modern LLMs decoder-only?
- 10 What is causal attention masking?
- 11 What is tokenization, and why does it matter?
- 12 How do BPE, WordPiece, and SentencePiece differ?
- 13 What is a context window, and what happens when you exceed it?
- 14 What is the KV cache, and why does it speed up inference?
- 15 What is perplexity?
- 16 What is Mixture of Experts (MoE)?
- 17 Parameters vs tokens vs embeddings: what's the difference?
- 18 What is temperature in LLM generation?
- 19 How do Top-K and Top-P sampling interact with temperature?
- 20 Base model vs instruction-tuned model: what changes?

PROMPT ENGINEERING

- 21 What is prompt engineering vs context engineering?
- 22 Zero-shot vs few-shot prompting: when to use each?
- 23 What is chain-of-thought (CoT) prompting?
- 24 How do reasoning models change your prompting strategy?
- 25 What are system prompts and role prompting?
- 26 What is prompt decomposition and prompt chaining?
- 27 How do you get reliable structured (JSON) output?
- 28 How do you version and regression-test prompts?
- 29 What is prompt injection (direct vs indirect)?
- 30 How do you defend an LLM app against prompt injection?

EMBEDDINGS & VECTOR DATABASES

- 31 What are embeddings?
- 32 How are text embeddings generated?
- 33 Embedding models vs LLMs: what's the difference?
- 34 Cosine similarity vs dot product vs Euclidean distance?
- 35 How do you choose an embedding model?
- 36 What is a vector database?
- 37 How does vector similarity search work?
- 38 What is Approximate Nearest Neighbor (ANN) search?
- 39 How do HNSW and IVF indexes work?
- 40 What is product quantization in vector search?
- 41 What factors affect vector search quality?
- 42 Metadata filtering: pre-filter vs post-filter?

04 · Retrieval-Augmented Generation

Q 43-60

43 What is Retrieval-Augmented Generation (RAG)?

44 Walk through the end-to-end RAG pipeline.

TOP

45 What is document chunking?

46 How do you choose chunk size and overlap?

TOP

47 Fixed vs recursive vs semantic chunking?

48 What problems does naive chunking cause?

49 What is hybrid search (dense + sparse)?

TOP

50 How does BM25 differ from vector search?

51 What is reranking, and why do cross-encoders help?

TOP

52 What are query rewriting and HyDE?

53 What is GraphRAG, and when is it worth it?

54 What is agentic RAG?

55 How would you design a production-grade RAG system?

TOP

56 How do you handle tables, images, and messy PDFs?

57 Long context vs RAG for very large documents?

58 How do you reduce irrelevant retrieval?

59 How do you prevent hallucinations in RAG?

TOP

60 How do you evaluate a RAG system?

TOP

05 - Fine-Tuning & Model Adaptation

Q 61-72

61 Prompting vs RAG vs fine-tuning: how do you decide?

TOP

62 What is supervised fine-tuning (SFT)?

63 What are RLHF and DPO, and how do they differ?

64 What is LoRA (Low-Rank Adaptation)?

65 What is QLoRA, and how does it save memory?

TOP

66 What is PEFT (Parameter-Efficient Fine-Tuning)?

67 Full fine-tuning vs LoRA: what are the trade-offs?

68 What is quantization?

69 GPTQ vs AWQ vs GGUF: 4-bit and 8-bit explained?

70 What is catastrophic forgetting, and how do you prevent it?

71 How do you prepare a high-quality fine-tuning dataset?

72 How do you evaluate a fine-tuned model?

RULE OF THUMB FOR Q61

Need **new knowledge**? Use RAG. Need a new **behavior, format, or tone**? Fine-tune. Try **prompting first**, because it's the cheapest to change.

06 - AI Agents & Tool Calling

Q 73-85

73 What is an AI agent?

74 How is an agent different from a chatbot?

75 What is tool / function calling?

TOP

76 How does an LLM decide when to call a tool?

77 What is the Model Context Protocol (MCP)?

78 Deterministic workflows vs autonomous agents?

TOP

79 What is the ReAct (Reason + Act) pattern?

80 How do agents manage memory and state?

81 How do you stop an agent from looping forever?

TOP

82 How do you design reliable multi-agent systems?

83 What are the major failure modes of AI agents?

TOP

84 How do you evaluate an AI agent?

85 How do you add human-in-the-loop (HITL) approval?

INTERVIEW TIP FOR Q78

Say this: use a **workflow** when the steps are known, and an **agent** only when the path can't be predicted. Interviewers want to hear you default to the simpler option.

07 - Evaluation & 08 - System Design

Q 86-100

GENAI EVALUATION

- 86 Why is evaluating LLM applications so hard? TOP
- 87 What is LLM-as-a-Judge, and what are its biases?
- 88 How do you detect and measure hallucinations?
- 89 Precision@K, Recall@K, MRR, and NDCG?
- 90 Faithfulness, answer relevance, and context precision?
- 91 How do you build a golden evaluation dataset?
- 92 Offline vs online evaluation (A/B tests, user feedback)?

PRODUCTION & LLM SYSTEM DESIGN

- 93 How would you design a production-grade GenAI app? TOP
- 94 How do you reduce LLM latency?
- 95 How do you reduce inference cost? TOP
- 96 How do you handle rate limits, timeouts, and provider outages?
- 97 Exact vs semantic caching: how do you implement it?
- 98 How do you monitor an LLM app in production? TOP
- 99 How do you scale RAG to millions of documents?
- 100 Design an end-to-end agent / RAG system: trade-offs, security, eval, cost, observability. TOP

Your 5-Week Study Plan

Q Roadmap

1 Transformers Week 1	2 Prompting Week 1	3 Embeddings Week 2	4 RAG Week 2-3
5 Fine-Tuning Week 3	6 Agents Week 4	7 Evaluation Week 4	8 System Design Week 5

THE 18 QUESTIONS TO MASTER FIRST

#44 RAG pipeline	#46 Chunk size & overlap
#49 Hybrid search	#51 Reranking
#55 Production RAG	#59 Stop hallucinations
#60 RAG evaluation	#61 Prompt vs RAG vs FT
#65 QLoRA	#75 Function calling
#78 Workflow vs agent	#81 Agent loops
#83 Agent failures	#86 Why eval is hard
#93 GenAI app design	#95 Cost reduction
#98 LLM monitoring	#100 End-to-end system

Learn these 18 deeply before skimming the other 82. Depth beats breadth in senior rounds.

"Design a production GenAI system from scratch."

STRUCTURE YOUR ANSWER IN 5 MOVES

1. Clarify requirements
2. Draw the architecture
3. Explain trade-offs
4. Show how you'd measure it
5. Estimate cost

COVER THESE 12 BUILDING BLOCKS

01 Ingestion Parse PDF/HTML, clean text, extract metadata	02 Chunking Semantic or recursive splits, tuned overlap
03 Embeddings Pick model by domain, dimensions, batch cost	04 Vector DB HNSW vs IVF, filters, sharding
05 Hybrid Retrieval Dense + BM25, fused with RRF	06 Reranking Cross-encoder on the top candidates
07 Prompting System rules, grounded context, citations	08 LLM + Tools Structured output, function calls, agent loop
09 Guardrails Input/output filters, injection defense	10 Caching Exact + semantic cache, streaming
11 Evaluation Faithfulness, relevance, golden sets	12 LLMOps Tracing, token budgets, rate limits, alerts

BEFORE YOU GO

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